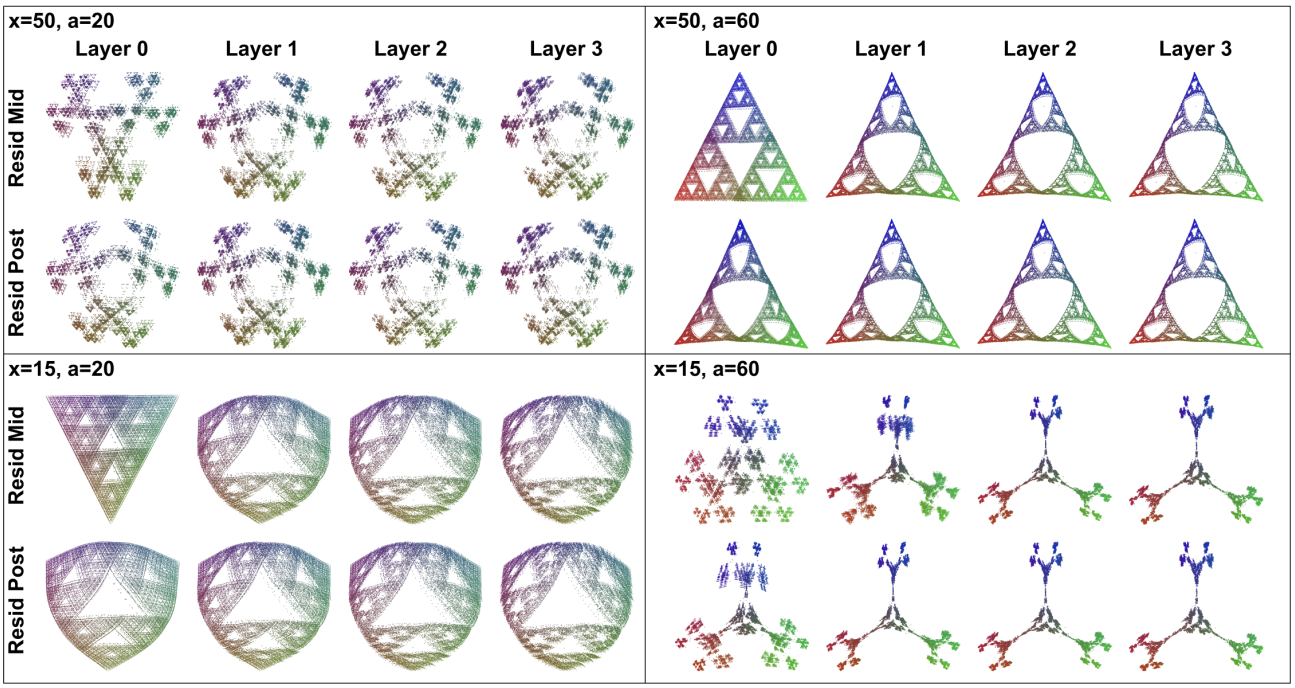

Figure A5: **Visualization of Activation Geometry via Regressions onto the Full Belief State.** The figure displays 2D projections of transformer activations after linear regression onto the Full (unconstrained) Belief state geometry (Eq. (2)) for all stages, arranged in a 2x2 grid. Each quadrant corresponds to a different experimental condition (combinations of x and a, labeled above each quadrant). Within each quadrant, the top row shows visualizations for Mid (Post-Attention/Pre-MLP) activations, and the bottom row shows visualizations for Post (Post-MLP) activations, across Layers 0-3 (columns). Points in all plots are colored according to the ground truth Full Belief state coordinates (Eq. (2)), projected into RGB space. Importantly, all visualizations show regressions to the Full Belief state geometry (and not the Constrained Belief geometry). Top Row (Mid Activations): Shows Mid activations after being regressed onto the Full Belief geometry (Eq. (2)). Bottom Row (Post Activations): Shows Post activations after being regressed onto the Full Belief geometry (Eq. (2)). This visualization qualitatively shows how well activations linearly map to the Full Belief state throughout the network. Notably, the residual stream activations after the first attention layer but before the MLP (Layer 0 mid, top-left plot within each quadrant), despite being regressed onto the Full Belief target, visually retain a structure resembling the Constrained Theory geometry (Eq. (5)). This indicates that the constrained structure is the dominant feature linearly recoverable from early Mid activations, even when seeking the best fit to the final target geometry. Comparing subsequent Mid and Post stages across layers (moving rightwards) reveals the accurate fit to the Full Belief state geometry.

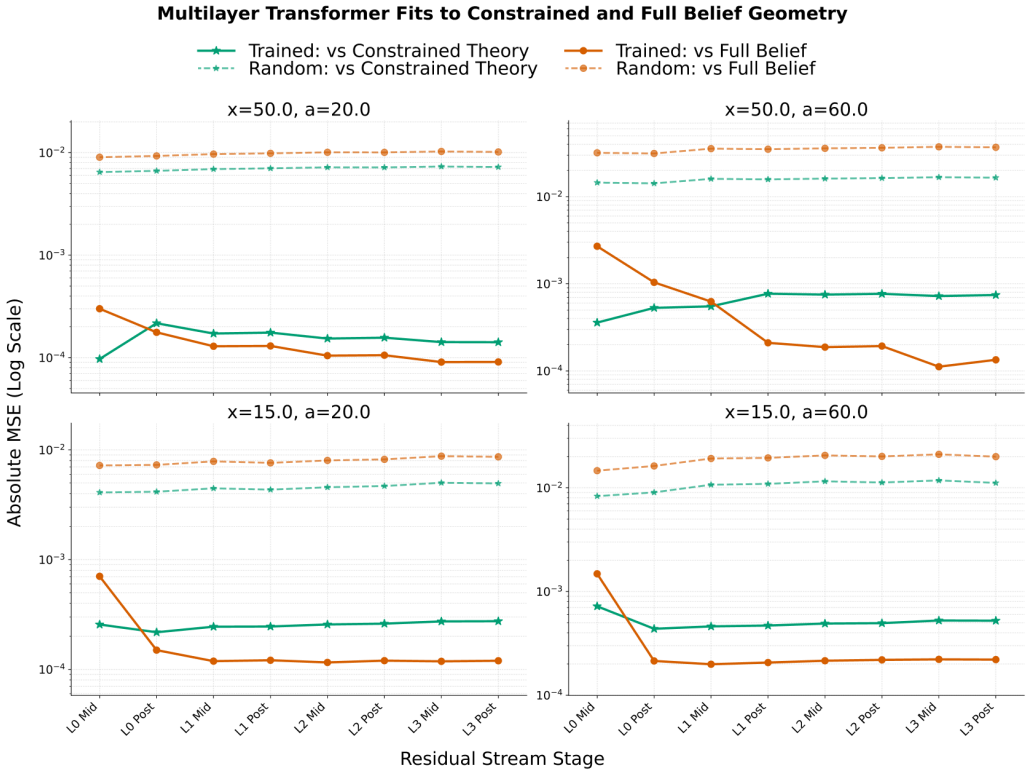

Figure A6: **Quantitative Fit of Multi-Layer Transformer Activations to Theoretical Geometries Across Residual Stream Stages.** Absolute Mean Squared Error (MSE, log scale) comparing activation representations to the Constrained Theory (Eq. (5)) and the Full (unconstrained) Belief state geometry (Eq. (2)) across interleaved residual stream stages (L0 Mid, L0 Post, ..., L3 Post) of a 4-layer transformer. The figure presents results for four experimental conditions (combinations of x=15, 50 and a=20, 60) in a 2x2 grid. Lines: Show absolute MSE comparing activation fits to theoretical geometries. Solid lines represent the trained model; dashed lines represent the random baseline. Green Stars: Fits of the residual stream activations to the Constrained Theory (Eq. (5)). Orange Circles: Fit of the residual stream activations to the Full Belief geometry (Eq. (2)). At the residual stream after the first attention but before the first MLP (L0 Mid), the fit to the Constrained Theory (green) is better (lower MSE) than the fit to the Full Belief geometry (orange) across all conditions, supporting the hypothesis that the initial layer's attention mechanism implements the constrained update. The fit to the Full Belief geometry (orange line) improves dramatically after the first MLP, and at that point the residual stream activations switch to fitting the Full Belief geometry better than the Constrained Belief geometry. The fit to the Constrained Theory (green line) does not show this convergence and may worsen in later layers. The MSE values for the trained transformer (solid lines) are consistently orders of magnitude lower than the corresponding random baselines (dashed lines), demonstrating that these geometric alignments are learned features resulting from training.

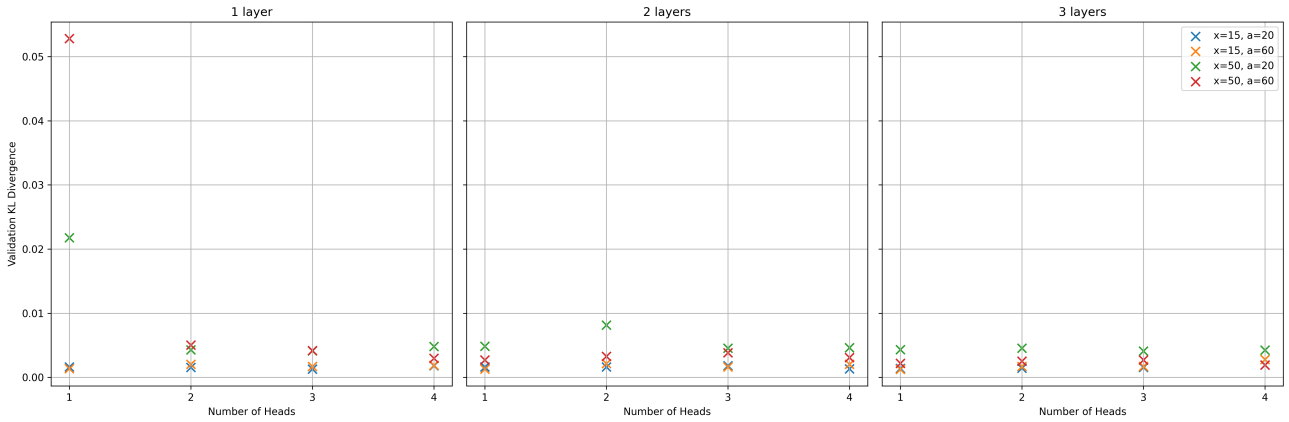

Figure A7: Validation KL divergence between model predictions and optimal probabilities across different architectural configurations. Results shown for various Mess3 parameter settings (x and α) and model architectures (number of heads and layers). The model achieves good performance with minimal architecture: a single layer with two attention heads is sufficient across parameter settings.

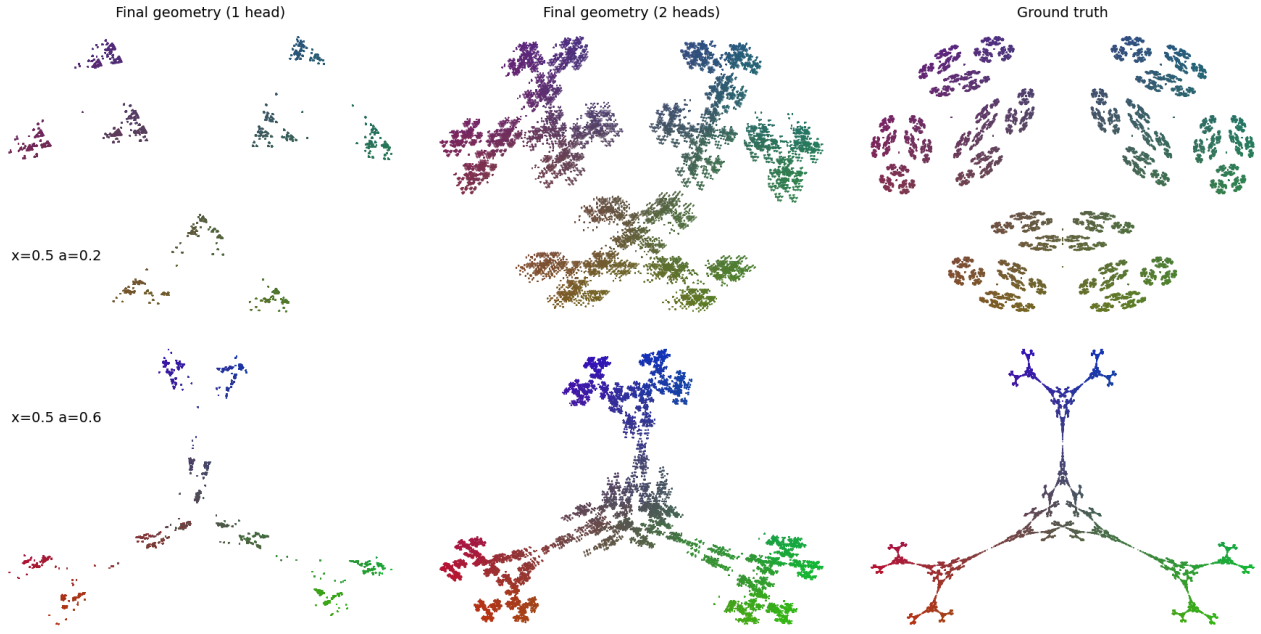

Figure A8: Comparison of learned belief geometry with one head (left) versus two heads (middle) against ground truth (right) for two different Mess3 parameter settings. With $x = 0.5$, where the optimal update pattern requires both positive and negative components, a single head fails to capture the correct geometry due to the non-negativity constraint of attention. Two heads allow the model to properly implement these updates, resulting in geometry that closely matches the ground truth.